

Kernels

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Motivation

- Most methods we've looked at so far are linear, which is limiting
- (Draw classifier with circle boundary)
- Consider the *feature map* $(x_1, x_2) \rightarrow (x_1^2, x_2^2)$
- (Draw points after the mapping)
- Data which is not linearly separable becomes linearly separable in the new space
- (Draw XOR example)
- Consider feature map $\phi: \mathbf{R}^2 \rightarrow \mathbf{R}^3$ where $\phi(x) = [x_1, x_2, x_1x_2]$
 - Then $\text{sign}(x_1x_2)$ works
- (Fail at drawing this in 3D)

Feature Maps

- A feature map ϕ maps a feature vector x to some higher-dimensional space
- Simple example: the padding trick
- $\phi(x) = [x, 1]$ and parameter vector $w = [p, b]$, both $\in \mathbf{R}^{d+1}$
- Before: classifiers like $\langle x, p \rangle > 0$
- After: classifiers like $\langle \phi(x), w \rangle = \langle x, p \rangle + b > 0$

Quadratic Feature Maps

- Instead of functions $x^T p + b > 0$, consider $x^T Q x + \sqrt{2} x^T p + b > 0$
 - $Q \in \mathbf{R}^{d \times d}, p \in \mathbf{R}^d, b \in \mathbf{R}$
 - Trust me on the $\sqrt{2}$ for now
- Note that $x^T Q x = \sum (x_i x_j) Q_{ij}$
- A dot product between “flattenings” of xx^T and Q , name $\overrightarrow{xx^T}$ and $\vec{Q}$ (draw)
- Take $\phi(x) = [\overrightarrow{xx^T}, \sqrt{2}x, 1]$ and $w = [\vec{Q}, p, b]$, both in $\mathbf{R}^{d^2+d+1}$
- Then $\langle \phi(x), w \rangle \leftrightarrow x^T Q x + \sqrt{2} x^T p + b$

Feature Maps

- Generally: instead of taking dot product of feature vector with parameter vector, map feature vector and take dot product with new parameter vector
- With quadratic feature map $\phi: \mathbf{R}^d \rightarrow \mathbf{R}^{d^2+d+1}$, computations go from $O(d)$ to $O(d^2)$
- What if we map to a very high-dimensional space?
- What if we have an infinite-dimensional feature map?
 - E.g., ($X = \mathbf{R}$) $\phi(x) = \left[1, x, \frac{x^2}{\sqrt{2!}}, \frac{x^3}{\sqrt{3!}}, \frac{x^4}{\sqrt{4!}}, \dots \right]$
 - Naively, can't be computed in finite time. What do?

Using Feature Maps in SVM (Dual)

$$\min_{\alpha \in \mathbf{R}^n, C \geq \alpha \geq 0} \frac{1}{2} \sum \sum \alpha^{(i)} \alpha^{(j)} y^{(i)} y^{(j)} \langle x^{(i)}, x^{(j)} \rangle - \sum_i \alpha^{(i)} \quad \text{s. t.} \quad \sum_i \alpha^{(i)} y^{(i)} = 0$$

Using Feature Maps in SVM (Dual)

$$\min_{\alpha \in \mathbf{R}^n, C \geq \alpha \geq 0} \frac{1}{2} \sum \sum \alpha^{(i)} \alpha^{(j)} y^{(i)} y^{(j)} \langle \phi(x^{(i)}), \phi(x^{(j)}) \rangle - \sum_i \alpha^{(i)} \quad \text{s.t.} \quad \sum_i \alpha^{(i)} y^{(i)} = 0$$

- Focus on quadratic $\phi(x) = [\overrightarrow{xx^T}, \sqrt{2}x, 1]$ for now
- How to compute $\langle \phi(x), \phi(x') \rangle$?
- Naively: compute $\phi(x)$, compute $\phi(x')$, take their dot product. $O(d^2)$ time
- BUT! We don't care about $\phi(x)$ or $\phi(x')$, only their dot product $\langle \phi(x), \phi(x') \rangle$

Quadratic Feature Map Kernel

- How to compute $\langle \phi(x), \phi(x') \rangle$?

$$\langle \phi(x), \phi(x') \rangle = \left\langle \overrightarrow{xx^T}, \overrightarrow{x'x'^T} \right\rangle + \left\langle \sqrt{2}x, \sqrt{2}x' \right\rangle + \langle 1, 1 \rangle$$

$$\left\langle \overrightarrow{xx^T}, \overrightarrow{x'x'^T} \right\rangle = \sum_{i,j} x_i x_j x'_i x'_j = \sum_{i,j} x_i x'_i x_j x'_j = \sum_i x_i x'_i \left(\sum_j x_j x'_j \right) = \langle x, x' \rangle^2$$

$$\langle \phi(x), \phi(x') \rangle = \langle x, x' \rangle^2 + 2\langle x, x' \rangle + 1 = (\langle x, x' \rangle + 1)^2$$

- Note that $(\langle x, x' \rangle + 1)^2$ can be computed in $O(d)$ time, instead of $O(d^2)$!
- $k(x, x') = (\langle x, x' \rangle + 1)^2$ is a *kernel*

Kernels

- $k: X \times X \rightarrow \mathbf{R}$ is a *kernel* if there exists a feature map ϕ such that $k(x, x') = \langle \phi(x), \phi(x') \rangle$
- $\phi(x)$ may be expensive (or impossible) to compute, but the kernel k may be tractable
 - Compare $O(d^2)$ for quadratic feature map versus $O(d)$ for kernel
- Polynomial kernel of degree t : $k(x, x') = (\langle x, x' \rangle + 1)^t$
- Gaussian/radial basis function: $k(x, x') = \exp(-\gamma \|x - x'\|_2^2)$
- Exponential dot product: $k(x, x') = \exp(\langle x, x' \rangle)$
 - $\exp(\langle x, x' \rangle) = 1 + \frac{\langle x, x' \rangle}{1!} + \frac{\langle x, x' \rangle^2}{2!} + \dots$. Feature map?

What makes a valid Kernel?

- First off, if you can construct a corresponding feature map ϕ
- This implies the following equivalent interpretation
- Let $x_1, \dots, x_n$ be an arbitrary dataset
- Let $K \in \mathbf{R}^{n \times n}$ be a matrix where $K_{ij} = k(x_i, x_j)$
- K is symmetric ($K_{ij} = K_{ji}$) and positive semidefinite
 - $v^T K v \geq 0$ for all vectors $v \in \mathbf{R}^n$
- The existence of a feature map implies these properties due to the Gram matrix (Expand $v^T K v$ on board)
 - Implication goes the other way too

Using Kernels in SVM (Dual)

Solve

$$\min_{\alpha \in \mathbf{R}^n, C \geq \alpha \geq 0} \frac{1}{2} \sum \sum \alpha^{(i)} \alpha^{(j)} y^{(i)} y^{(j)} K_{ij} - \sum_i \alpha^{(i)} \quad \text{s.t.} \quad \sum_i \alpha^{(i)} y^{(i)} = 0$$

- $K_{ij} = \langle \phi(x^{(i)}), \phi(x^{(j)}) \rangle = k(x^{(i)}, x^{(j)})$
- How to classify new point?
 - $w = \sum \alpha^{(i)} y^{(i)} \phi(x^{(i)})$, but can't compute $\phi(x^{(i)})$
 - $\text{sign}(\langle w, \phi(x) \rangle + b) = \text{sign}(\langle \sum \alpha^{(i)} y^{(i)} \phi(x^{(i)}), \phi(x) \rangle + b) = \text{sign}(\sum \alpha^{(i)} y^{(i)} k(x^{(i)}, x) + b)$
- Primal representation: $O(D)$ storage, $O(D)$ inference time, D is $\dim(\phi(x))$
- Dual representation: construct K : $O(n^2 c_k)$, store K : $O(n^2)$, $O(sc_k)$ inference time
 - s support vectors, c_k cost of one kernel evaluation